



Atom Computing's next-generation quantum computing system, currently in development in their commercial operations facility in Boulder, CO (Courtesy: Atom Computing)

which was technically open source, but not open source in spirit.

For example, Meta released a version of Llama 2 under a so-called open-source license, but under the condition that if you have more than 700,000 monthly users, you need to approach the company for a license. Red Hat has stopped releasing its enterprise changes to the public. HashiCorp, which makes software like Terraform, Vault and Packer, changed its licensing from the Mozilla Public License (MPL) to the partially open Business Source License (BSL), which restricts users from using HashiCorp's products to create software that will compete with the company's offerings!

Forrester terms such licensing ploys, which are only partially open source as 'ajar source' and they predict that more corporates will be releasing software as ajar source in 2024.

Industry cloud platforms (ICPs). According to a Gartner report, ICPs address industry-relevant

business outcomes by combining underlying software-as-a-service (SaaS), platform-as-a-service (PaaS), and information-as-a-service (IaaS) services into a whole product offering with composable capabilities, and more than 70% of enterprises will use ICPs by 2027, up from less than 15% in 2023.

Agri-tech for sustainable food production. Technology is helping optimise the farm-to-fork lifecycle by increasing yield, optimising costs, and reducing wastage. We read about several of these solutions in the April 2023 issue of Electronics For You.

"Climate issues, geopolitical tensions, water, and energy shortages, rising fertiliser costs, and inefficient production methods exert pressure on agricultural production. AgTech solutions can help improve crop yields, use equipment and livestock efficiently, plan harvests better, and adopt sustainable agrifood-production methods," explains a Deloitte report. It predicts that there will be nearly 300 million installed IoT

endpoints by the end of 2024, for purposes ranging from precision crop farming and livestock management to agricultural equipment tracking. The overall agricultural technology revenue opportunity—including these IoT end points and connectivity devices—will be \$18 billion globally in 2024, representing a 19% compound annual growth rate (CAGR) between 2020 and 2024.

Telco trends—Looking skywards

Direct-to-device (D2D) satellite connectivity. Unlike incumbents like geostationary orbit (GEO) and low Earth orbit (LEO) constellations (such as Starlink, OneWeb, and Amazon's Kuiper) that need a dish or other large receiving equipment, D2D connectivity is provided directly to the user's mobile phone or any standard mobile device.

Iridium Communications partnered with Qualcomm to successfully develop and demonstrate D2D tech using Iridium's satellite network and Qualcomm's Snapdragon mobile platform CPU. However, since no smartphone manufacturers included the tech in their devices, Qualcomm chose to terminate the agreement in December 2023.

Speaking to the media, Iridium CEO Matt Desch had remarked, "While I'm disappointed that this partnership didn't bear immediate fruit, we believe the direction of the industry is clear toward increased satellite connectivity in consumer devices... Led by Apple today, MNOs and device manufacturers still plan, over time, to provide their customers with expanded coverage and new satellite-based features." SpaceX backed Starlinks proposes to launch SMS services for its direct-to-cell solution in 2024, and voice, data, and IoT services in 2025.